

Additional "Simple" Mohr's Circle Problems

① Given a center: $I_{center} = 32 \text{ in}^4$

Plus: Axes $P \& Q$ are $\perp$.

And: $I_P = 20 \text{ in}^4$.

Find: I_Q

Principle to use: $I_x + I_y = I_{max} + I_{min} = I_P + I_Q$
for any set of $\perp$ axes.

$$I_{center} = I_{AVG} = \frac{I_P + I_Q}{2} = 32 \text{ in}^4$$

$$I_P + I_Q = 64$$

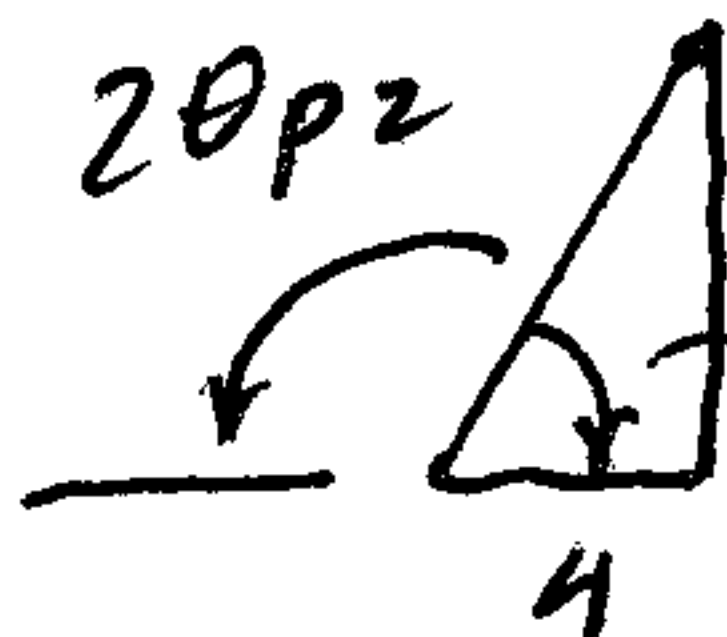
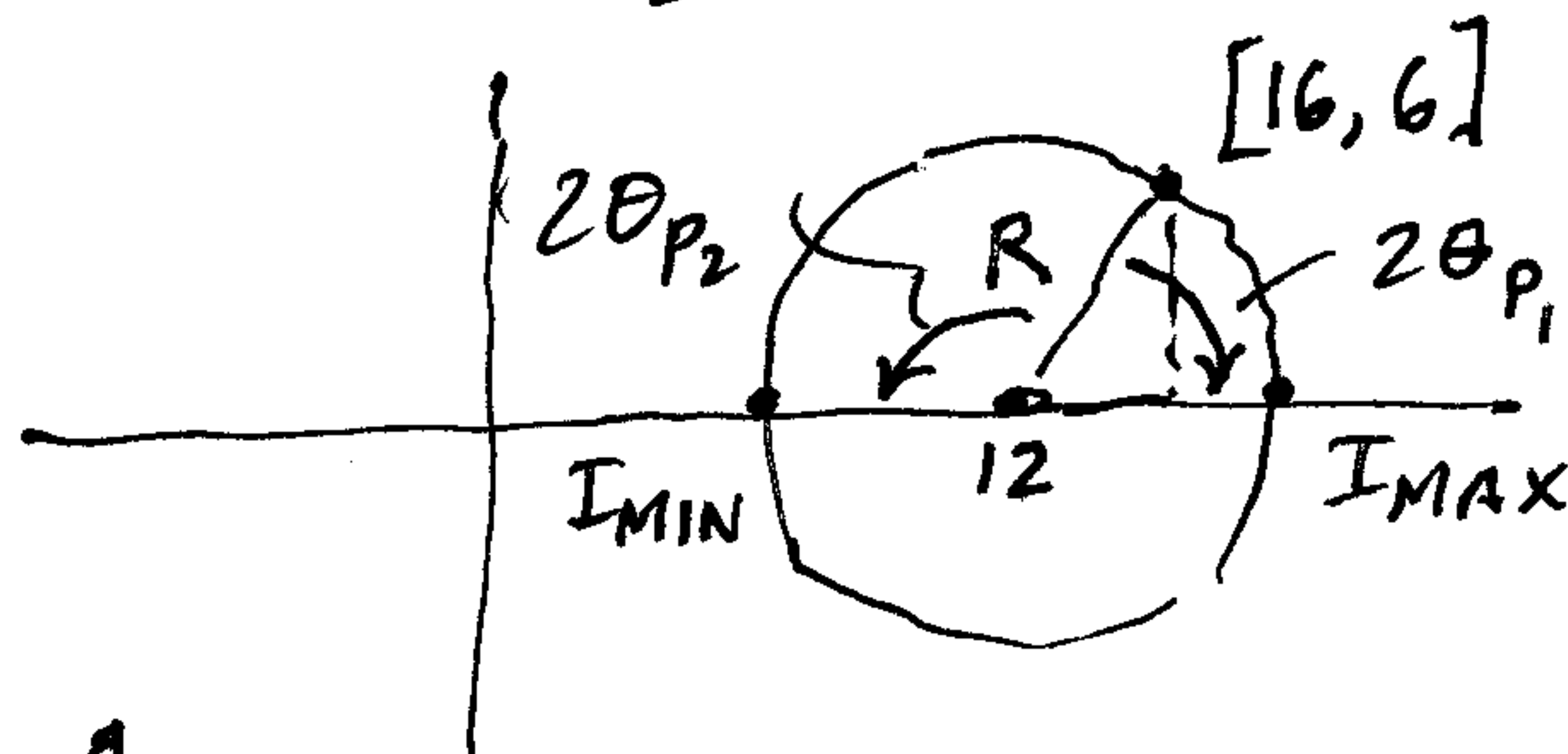
20

Thus, $I_Q = 44 \text{ in}^4$

② For a set of axes $P-Q$, $I_P = 16 \text{ in}^4$
and the POI $I_{PQ} = 6 \text{ in}^4$.

$$I_{center} = 12 \text{ in}^4$$

Find the angles from
the P axis to the
 I_{max} ("major" axis)
and I_{min} ("minor" axis)



$$2\theta_{P1} = \tan^{-1}(6/4)$$

$$2\theta_{P1} = 56.3^\circ$$

$$\theta_{P1} = 28.2^\circ$$

$$2\theta_{P2} = 180 - 56.3 = 123.7$$

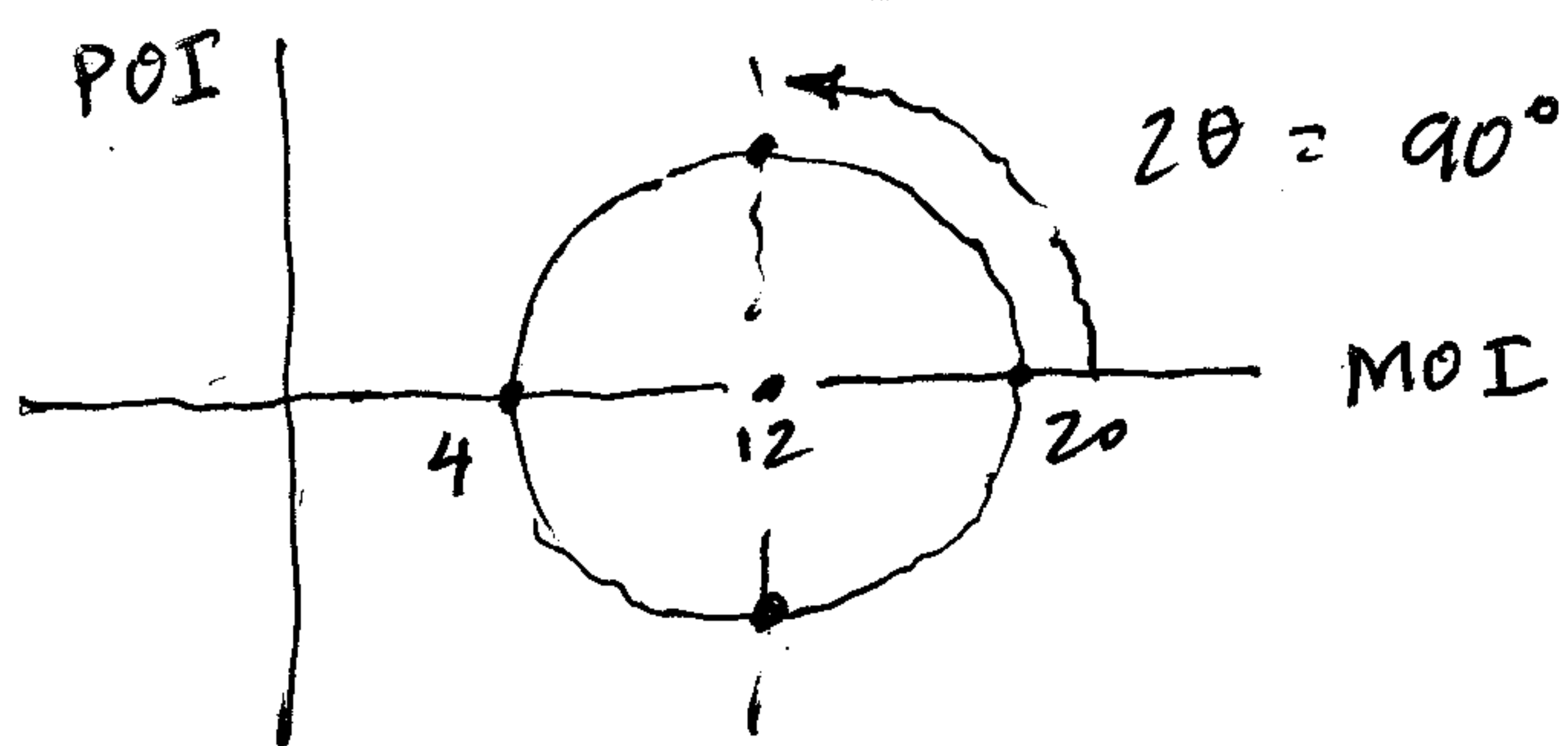
$$\theta_{P2} = 61.9^\circ$$

③ Given:

• Dia of MC = 8 in^4

• $I_{\text{CENTER}} = 12 \text{ in}^4$

Find POI about a pair of $\perp$ axes rotated 45° from the principal axes on the original area.



* A rotation of 45° on the element is $2(45) = 90^\circ$ on MC.

Thus, POI = Radius = 8 in^4